Learning Objective 6.5: I can implement FHSS transmission and reception, including hop-sequence synchronization.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **The hop synthesizer.** A hopper like the L16 transmitter runs at $f_s = 128$ kHz. A Vector Source lists the hop frequencies, a Repeat block holds each one for one hop, and a VCO (complex) with sensitivity 2π turns each value, in hertz, into a tone. The hopper uses 6 channels spaced 12 kHz apart and centered on 0 Hz, and each hop lasts 125 ms.

   (a) Find the frequency of each channel, $k = 0$ to 5.

   Centered on 0 Hz, channel k sits at

   $$f_k = 12\text{ kHz} \times (k - 2.5)$$

   which gives -30, -18, -6, $+6$, $+18$, and $+30$ kHz.

   $\pm 6, \pm 18, \pm 30$ kHz

   (b) Find the Repeat block's interpolation, and the time for one cycle that visits every channel once.

   $$128{,}000 \times 0.125\text{ s} = 16{,}000\text{ samples}$$
   $$6 \times 125\text{ ms} = 750\text{ ms}$$

   Repeat: 16,000

   cycle: 750 ms

   (c) The FSK signal on each hop is about 4 kHz wide. In words, explain why a channel spacing of 3 kHz would not work, and what the receiver's low-pass filter after de-hopping does with the other channels at 12 kHz.

   At 3 kHz spacing each 4 kHz-wide signal would overlap its neighbors, so a receiver tuned to one channel would also pick up energy from the next one, and a jammer on one channel would spill into two. At 12 kHz the channels do not overlap: after de-hopping, the wanted hop sits at 0 Hz and every other channel sits 12 kHz or more away, outside a low-pass filter of about ± 2 kHz, which removes them.

2. **Hop timing.** A hopper sends one packet per hop, as the L16 link does: 2 guard bytes, the 12-byte access code and header, a payload of N bytes, and 2 more guard bytes. It runs at $R_s = 2000$ baud, with 16 channels.

   (a) For a 26-byte payload, find the bits per hop, the time on one hop, and the hop rate.

$$n = 8 \times (2 + 12 + 26 + 2) = 336 \text{ bits}$$
$$T_{\text{hop}} = \frac{336}{2000 \text{ baud}} = 168 \text{ ms}$$
$$\text{hop rate} = \frac{1}{0.168 \text{ s}} = 5.95 \text{ hops/s}$$

   bits: 336

   hop: 168 ms

   rate: 5.95 hops/s

   (b) Find the time for one cycle of the 16 channels, and the share of each hop that carries the payload.

$$T_{\text{cycle}} = 16 \times 168 \text{ ms} = 2.69 \text{ s}$$
$$\frac{8 \times 26}{336} = 0.62 = 62\%$$

   cycle: 2.69 s

   payload: 62%

   (c) How long is each guard, in milliseconds, and roughly what range of receiver timing error should it absorb?

   Each guard is 16 bits, $16/2000 = 8$ ms, so the receiver can change channel about 8 ms early or 8 ms late and still catch the access code, header, and payload.

   about ± 8 ms

   (d) In words, explain why the hopper makes each packet exactly one hop long instead of letting packets run across hop boundaries.

   If a packet ran across a boundary, part of it would be on one channel and part on the next, so a jammer on either channel, or a receiver whose timing is a little off, would lose the whole packet. With one packet per hop the boundaries fall in the guard bits between packets, a lost hop costs exactly one packet, and the receiver can count packets to check it is on the right hops.

3. **The FCC dwell rule.** FCC Part 15.247 lets a narrowband hopper at 915 MHz transmit without a license if it hops over at least 50 channels, uses every channel equally, and stays on any one channel no more than 0.4 s in any 20 s. The L16 link hops over 8 channels with 240 ms hops, each channel once per cycle.

   (a) Find how long the L16 link spends on one channel in 20 s.

   The link makes $20/0.240 = 83.3$ hops in 20 s, and each channel gets one in 8:

   $$\frac{83.3}{8} \times 0.240\ \text{s} = 2.5\ \text{s}$$

   which is more than six times the 0.4 s limit.

   2.5 s

   (b) Show that, with every channel used equally, the time on one channel in 20 s is $20\ \text{s}/N$ whatever the hop length, and find the smallest N that meets the 0.4 s limit.

   With hops of length T, there are $20/T$ hops in 20 s and each channel gets $1/N$ of them, so the time on one channel is

   $$\frac{20}{T} \cdot \frac{1}{N} \cdot T = \frac{20\ \text{s}}{N} \leqslant 0.4\ \text{s} \quad \Rightarrow \quad N \geqslant 50$$

   $N \geqslant 50$

   (c) The rule also limits each single visit: a hop longer than 0.4 s breaks it even with 50 channels. In words, explain why a permutation sequence, like the L16 link's, makes the equal-use rule easy to meet, and what a sequence drawn independently at random each hop would do instead.

   A permutation visits every channel exactly once per cycle, so every channel gets exactly the same time in every cycle, and the average use is equal by construction. A channel drawn at random each hop is equal only on average: over a short window some channels come up more often than others, and one could exceed 0.4 s in a 20 s window even when the long-run average is fine. It could also repeat a channel on consecutive hops.

4. **Jamming.** A hopper uses 20 channels, each equally. A jammer transmits a strong tone on each of 3 channels, strong enough that any hop on those channels is lost. On a channel it covers, the jammer needs a power at least equal to the signal's to take the hop, as in the L16 jammer sweep.

   (a) Find the fraction of packets delivered.

   $$P = 1 - \frac{3}{20} = 0.85 = 85\%$$

   85%

   (b) To take every hop, the jammer must put at least the signal's power on all 20 channels. How much more total power, in dB, does that take than jamming one channel?

   $$10 \log_{10} 20 = 13 \text{ dB}$$

   13 dB

   (c) The jammer spreads its 3-channel power evenly over all 20 channels instead. Is any hop lost? Find the jammer-to-signal ratio on each channel.

   Each of the 3 tones had at least the signal's power, so their total is at least 3 times it. Spread over 20 channels, each gets $3/20$ of the signal's power:

   $$10 \log_{10}(3/20) = -8.2 \text{ dB}$$

   That is below the 0 dB a jammer needs, so no hop is lost.

   none; -8.2 dB

   (d) Bluetooth's adaptive frequency hopping lets the receiver mark channels that keep losing packets as bad and drop them from the sequence. In words, explain what happens to the delivered fraction when this hopper drops the 3 jammed channels, and why the jammer cannot simply follow.

   With the 3 jammed channels dropped, the hopper uses the other 17, none of which is jammed, so every packet arrives again. The jammer does not know the sequence, so it cannot tell which channels are now in use except by listening; to hit the hops it would have to move to new channels and cover them too, and then the receiver drops those. Each channel the hopper drops makes it a little easier to find, which is why Bluetooth keeps at least 20.

5. **Synchronization.** The L16 link hops over 8 channels with key 448, in the order 7, 2, 0, 6, 4, 1, 3, 5, with 240 ms hops and 16 guard bits, 16 ms, at each end of every hop. Its receiver parks on the sync channel, channel 7, until hop 0 arrives there.

   (a) A receiver with the right timing tunes one hop behind the transmitter. List the transmitter's and the receiver's channels for hops 1 to 4, and find how many packets arrive in a full cycle.

   0

   On hop k the transmitter is on entry k of the list and the receiver on entry $k-1$: hop 1 is 2 and 7, hop 2 is 0 and 2, hop 3 is 6 and 0, hop 4 is 4 and 6. Consecutive entries of a permutation are never equal, so they never match, and no packet arrives in any cycle.

   (b) A receiver starts and parks on channel 7 at 0.70 s after the start of a hop 0. How long does it wait before it locks?

   1.22 s

   The next hop 0 starts one cycle after the last, at $8 \times 0.240 = 1.92$ s:

   $$1.92 - 0.70 = 1.22 \text{ s}$$

   (c) After locking, the receiver changes channel 10 ms late on every hop. A teammate's receiver changes channel 20 ms early. Explain what each one misses and whether its packets arrive.

   The late receiver spends the first 10 ms of each hop on the previous channel, so it misses 10 of the 16 leading guard bits; the access code, header, and payload all arrive, and so does every packet. The early receiver leaves 20 ms before the hop ends, so it misses the 16 trailing guard bits and the last 4 bits of the newline; the payload is wrong, and almost no packet arrives intact.

   (d) In words, explain why parking on one channel tells the receiver which hop the transmitter is on, and what would go wrong if the sequence visited channel 7 twice in each cycle.

   The sequence is a permutation, so the transmitter is on channel 7 only on hop 0; power appearing there means hop 0 has just started, which gives both the hop number and the hop boundary. If channel 7 appeared twice, the receiver could not tell which of the two visits it heard, and half the time it would lock to the wrong hop number, off by several hops, and receive nothing until it parked again.

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